

## LIVER

LA (18:2n-6)

FADS1/6, ELOVL2/5

Arachidonic Acid (20:4n-6)

PLA2G6/7, PLD1/2

Free AA



PGE / PGD / LTs / HETEs / EETs / 20-HETE

## SKELETAL MUSCLE

PORTAL / SYSTEMIC  
CIRCULATION

EP4 (PTGER4) / TP (TBXA2R)  
CB4R — Membrane GPCRs

Eicosanoids as endocrine signals  
↓  
cAMP-PKA / Ca<sup>2+</sup> -PKC  
Second Messengers

PPARα / PPARδ — Nuclear Receptors  
(EET as ligand)

FOXO3 — Transcription Factor  
(r = 0.841 with CBR1)

TRIM63 / FBXO32  
MuRF1 / Atrogin-1

Ubiquitin-Proteasome  
System Activation

↑ Muscle Protein Degradation  
↓ Protein Deposition

CBR2 ↔ TRIM63: r = 0.927, P = 0.0009

CYP2E1 ↔ PPARA: r = 0.884, P = 0.0036

**Key Cross-Tissue Correlations**

GPX3 ↔ TRIM63: r = 0.839, P = 0.0092